Yen-Shi Wang

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Dedicated C++ programmer with deep learning inference, compilers, performance optimization, and LLM background

Experience

NVIDIA — TensorRT Santa Clara, CA

Senior Deep Learning Software Engineer — Data Centers Jul. 2022 - Present

- Implemented multi-device TensorRT runtime in C++ using NCCL backend to support inference with multiple nodes.
- Wrote TensorRT v9.1.0 NeMo demo from scratch in Python, which supports FP8 inference on GPT-3 style LLM network.
- Added FP8 inference support for NeMo and profiled demo runtime and improved generation decoding latency by 30%.

Deep Learning Software Engineer — Automotive Safety Mar. 2021 - Jun. 2022

- Point of contact for TensorRT Consistency Checker. Implemented deconvolution node validation from scratch.
- Improved all safety layer tests, performed code inspection, and went through verification to confrom ISO 26262 safety standard.
- Debug C++ code with GDB on a weekly basis and fix software bugs caused by use after free, random failures, and type errors.

Performance Software Engineering Intern May. 2020 - Aug. 2020

- Improved C++ multithreading server for MLPerf Inference BERT benchmark to scale linearly from 1- to 20-GPU machines.
- Optimized GPU utilization with CUDA streams and graphs, solved runtime bugs on CPU and GPU, boosted throughput by 25%.

Skymizer Taipei, Taiwan

C++ Developer — worked on Open Neural Network Compiler (ONNC) Apr. 2019 - Jul. 2019

- Rewrote 21 optimizations for deep learning ONNX models, added testing framework from scratch, and ported into ONNC.
- Initiated quantization flow in ONNC backend to perform 8-bit quantization for NVIDIA Deep Learning Accelerator (NVDLA).

BravoAI Co., Ltd. Taipei, Taiwan

Software Engineer — focused on Optical Character Recognition Mar. 2018 - Sep. 2018

- Developed a system using PyTorch and TensorFlow to convert fields on medical certificate from paper into electronic forms.
- Deployed entire system with four Docker containers running Flask web service, operating 0.5 image/sec at over 95% accuracy.

Skills

Programming Languages	C++, C, Python, Bash, Java, Javascript
Automations and Databases	Docker, Terraform, Ansible, Kubernetes, MySQL, MongoDB
Others	Linux, Git, GDB, CMake, LLVM, CUDA

Education

Carnegie Mellon University Pittsburgh, PA

Master of Science in Electrical and Computer Engineering, GPA: 3.91/4.0 Dec. 2020

Coursework: Compilers Design, Optimizing Compilers, Cloud Computing, Storage Systems

National Taiwan University Taipei, Taiwan

Bachelor of Science in Computer Science and Information Engineering, GPA: 3.85/4.0 Jan. 2019

Bachelor of Science in Electrical Engineering, GPA: 3.85/4.0 Jan. 2019

Coursework: Operating Systems, System Programming, Deep Learning, Multimedia Analysis

Projects

C0 Compiler (project lasts for entire Compilers Design at CMU) Sep. 2020 - Dec. 2020

- Implemented frontend parser, type checker, SSA optimizations on IR, x86 code generation, register spilling for C0 language.
- Generated code is 40% faster than GCC -O0, and included conditional constant propagation and tail call optimization.
- Modified frontend to support new intrinsic vector type based on C0, and emit SIMD instructions on vector<int> to speed up.

Distinctness Analysis in LLVM for C/C++ (final project in Optimizing Compilers at CMU) Mar. 2020 - May. 2020

- Created an LLVM Module Pass to generate function call graphs and perform Andersen's pointer analysis.
- Read LLVM doxygen, grew familiar with LLVM Infrastructure and dealt with Functions, Loops and at least 10 Instructions.

Honors

2018 **Rank 116**, Google Code Jam 2018, Round 1C

2017 **Silver Medal**, ACM-ICPC Asia Hua-Lien Regional Contest Hua-Lien, Taiwan

2013 **Silver Medal**, 54th International Mathematical Olympiad (IMO) Santa Marta, Colombia